

Lecture 8: The Matrix (continued)

Instructors:

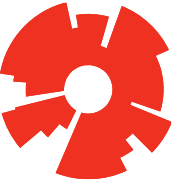
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[Based on slides for „Decoding the Matrix“]



WEEK

FOUR



Agenda

Chapter 4

Lecture 8:

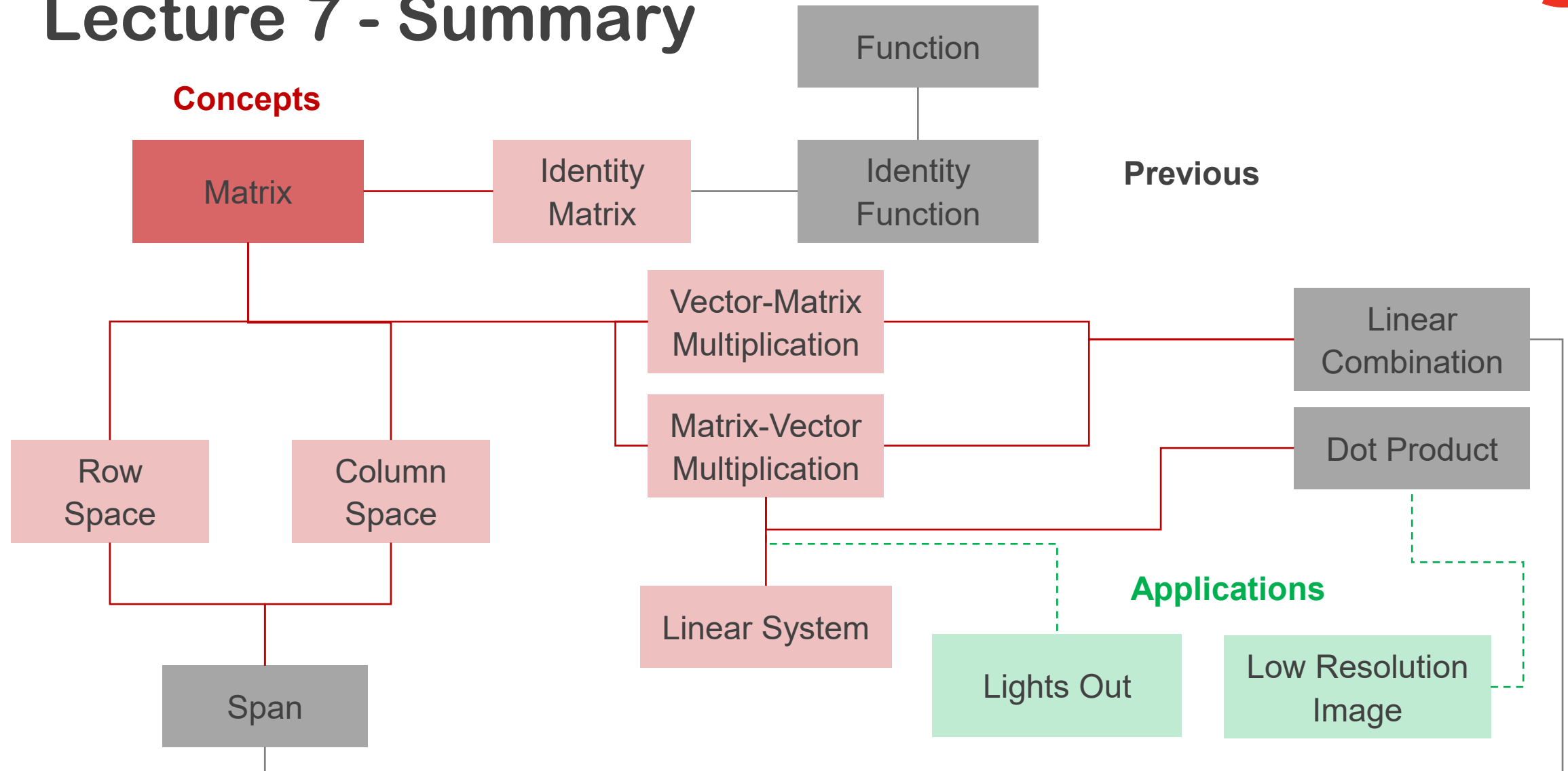
Null Space of a Matrix
Error-Correcting Codes
From Matrix to Function
From Function to Matrix
Matrix Scaling
Matrix applied to Standard Generators





Lecture 7 - Summary

Concepts

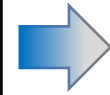




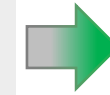
Recap: Matrix-Vector Equation

$$\begin{cases} x_1 + 0.5x_2 + 2x_3 + 4x_4 = 27 \\ 3x_2 + 3x_3 + 2x_4 = 31 \\ x_3 + 5x_4 = 13 \\ 2x_4 = 4 \end{cases}$$

System of LE



$$\begin{bmatrix} 1 & 0.5 & 2 & 4 \\ 0 & 3 & 3 & 2 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 2 \end{bmatrix} \cdot \mathbf{x} = \begin{bmatrix} 27 \\ 31 \\ 13 \\ 4 \end{bmatrix}$$



$$\overset{A}{\begin{bmatrix} 1 & 0.5 & 2 & 4 \\ 0 & 3 & 3 & 2 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 2 \end{bmatrix}} * \mathbf{x} = \overset{b}{\begin{bmatrix} 27 \\ 31 \\ 13 \\ 4 \end{bmatrix}}$$

Matrix-Vector Equation

We can go from a System of LE to...



... a Matrix-Vector Equation or vice versa



Null Space of a Matrix (I)

Definition: Null space of a matrix A is $\{\mathbf{u} : \underline{A * \mathbf{u} = \mathbf{0}}\}$. Written $Null A$

Example:

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 9 \end{bmatrix} * [0, 0, 0]^T = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

so the null space includes $[0, 0, 0]^T$

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 9 \end{bmatrix} * [6, -1, -1]^T = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

so the null space includes $[6, -1, -1]^T$

So, the **Null Space** is the set of vectors that when multiplied with a matrix lead to the zero vector - like in a homogeneous LS





Null Space of a Matrix (II)

We just saw:

Null space of a matrix $\begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \end{bmatrix}$ (e.g., the first row reflects the first linear equation)

equals the solution set of the homogeneous linear system

$$\begin{array}{rcl} \mathbf{a}_1 \cdot \mathbf{x} & = & 0 \\ & \vdots & \\ \mathbf{a}_m \cdot \mathbf{x} & = & 0 \end{array}$$

We just saw that a matrix-vector equation can represent a linear system, and now this?!





Null Space of a Matrix (III)

This shows: *Null space of a matrix is a vector space.*

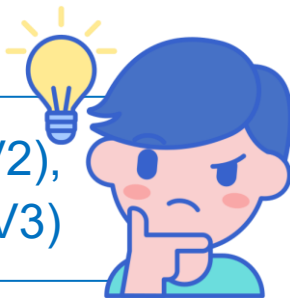
Can also show it directly, using algebraic properties of matrix-vector multiplication:

Property V1: Since $A * \mathbf{0} = \mathbf{0}$, the null space of A contains $\mathbf{0}$

Property V2: if $\mathbf{u} \in \text{Null } A$ then $A * (\alpha \mathbf{u}) = \alpha (A * \mathbf{u}) = \alpha \mathbf{0} = \mathbf{0}$ so $\alpha \mathbf{u} \in \text{Null } A$

Property V3: If $\mathbf{u} \in \text{Null } A$ and $\mathbf{v} \in \text{Null } A$
then $A * (\mathbf{u} + \mathbf{v}) = A * \mathbf{u} + A * \mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0}$
so $\mathbf{u} + \mathbf{v} \in \text{Null } A$

So, if u is a solution, then e.g., a multiple of u is a solution as well (V2),
and if u and v are solutions, then they together are also a solution (V3)





Null Space of a Matrix (IV)

Proposition: Null space of a matrix is a vector space.

(... and a vector space can be represented via a span)

Example:

$$\text{Null} \begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 9 \end{bmatrix} = \text{Span} \left\{ \begin{bmatrix} 6 \\ -1 \\ -1 \end{bmatrix} \right\}$$

How can we determine this?
Remind me!





Null Space of a Matrix (V)

Example: $\text{Null} \begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 9 \end{bmatrix} = \text{Span} \left\{ \begin{bmatrix} 6 \\ -1 \\ -1 \end{bmatrix} \right\}$ We don't know this yet

Definition: Null space of a matrix A is $\{\mathbf{u} : \underline{A * \mathbf{u} = \mathbf{0}}\}$. Written $\text{Null } A$

2 equations, 3 unknowns

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 9 \end{bmatrix} * [u_1, u_2, u_3]^T = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \longrightarrow \quad \begin{cases} u_1 + 2u_2 + 4u_3 = 0 \\ 2u_1 + 3u_2 + 9u_3 = 0 \end{cases}$$

We already know the solution cannot be a point
(**unique solution**)

No unique
solution
possible ...



Null Space of a Matrix (VI)

1. Solve the (homog.) System of LE

2 equations, 3 unknowns

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 9 \end{bmatrix} * [u_1, u_2, u_3]^T = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Rightarrow \quad \left| \begin{array}{l} u_1 + 2u_2 + 4u_3 = 0 \\ 2u_1 + 3u_2 + 9u_3 = 0 \end{array} \right|_{R_2 - 2R_1} \quad \Leftrightarrow \quad \left| \begin{array}{l} u_1 + 2u_2 + 4u_3 = 0 \\ -u_2 + u_3 = 0 \end{array} \right|$$

$$\Leftrightarrow \left| \begin{array}{l} u_1 + 2(u_3) + 4u_3 = 0 \\ u_2 = u_3 \end{array} \right| \quad \Leftrightarrow \quad \left| \begin{array}{l} u_1 = -6u_3 \\ u_2 = u_3 \end{array} \right|$$

2. State Solution Set in terms of the Span

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -6u_3 \\ u_3 \\ u_3 \end{bmatrix} = u_3 \begin{bmatrix} -6 \\ 1 \\ 1 \end{bmatrix} \quad \Rightarrow \quad \text{Span} \left\{ \begin{bmatrix} -6 \\ 1 \\ 1 \end{bmatrix} \right\}$$

Solution Set

Apart from the
solution set, it is

$$\text{Null} \begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 9 \end{bmatrix}$$



Null Space of a Matrix (VII)

Proposition: Null space of a matrix is a vector space.

Example:

$$\text{Null} \begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 9 \end{bmatrix} = \text{Span} \left\{ \begin{bmatrix} -6 \\ 1 \\ 1 \end{bmatrix} \right\}$$

If $[6, -1, -1]^T$ is a solution, then e.g., $2.5 * [6, -1, -1]^T = [15, -2.5, -2.5]^T$ is also a solution **[V2]**

And if they are solutions, then $[6, -1, -1]^T + [15, -2.5, -2.5]^T = [21, -3.5, -3.5]^T$ is also a solution **[V3]**

And they are all in the $\text{Span} \left\{ \begin{bmatrix} -6 \\ 1 \\ 1 \end{bmatrix} \right\}$

(... and we already established the span is in general a vector space)



Null Space of a Matrix ($\mathcal{V}$)

Earlier, we saw:

If $\mathbf{u}_1$ is a solution to the linear system

$$\begin{array}{rcl} \mathbf{a}_1 \cdot \mathbf{x} & = & \beta_1 \\ & \vdots & \\ \mathbf{a}_m \cdot \mathbf{x} & = & \beta_m \end{array}$$

then the solution set is $\mathbf{u}_1 + \mathcal{V}$,

where $\mathcal{V}$ = solution set of

$$\begin{array}{rcl} \mathbf{a}_1 \cdot \mathbf{x} & = & 0 \\ & \vdots & \\ \mathbf{a}_m \cdot \mathbf{x} & = & 0 \end{array}$$

Restated: If $\mathbf{u}_1$ is a solution to $A * \mathbf{x} = \mathbf{b}$ then solution set is $\mathbf{u}_1 + \mathcal{V}$
where $\mathcal{V} = \text{Null } A$

We have seen this in a different context before (now here: for **matrix-vector multiplication**)



Null Space of a Matrix (VI)

Remember
(some lectures
ago)

If $\mathbf{u}_1$ is a solution to a linear system then

$$\{\text{solutions to linear system}\} = \{\mathbf{u}_1 + \mathbf{v} : \mathbf{v} \in \mathcal{V}\}$$

where $\mathcal{V} = \{\text{solutions to corresponding homogeneous linear system}\}$

Restated: If $\mathbf{u}_1$ is a solution to $A * \mathbf{x} = \mathbf{b}$ then solution set is $\mathbf{u}_1 + \mathcal{V}$
where $\mathcal{V} = \text{Null } A$



System of LE: Linear System LE $\rightarrow$ Homogeneous LE $\rightarrow$ Find Span



Matrix-Vector Eq.:

$$A\mathbf{x} = \mathbf{b}$$

$\rightarrow$

$$A\mathbf{x} = \mathbf{0}$$

$\rightarrow$

Find Null A

(Later, we'll have a different way of finding *Null A* than finding *Span*)



Null Space of a Matrix (VII)

Proposition: If $\mathbf{u}_1$ is a solution to $A * \mathbf{x} = \mathbf{b}$ then solution set is $\mathbf{u}_1 + \mathcal{V}$
where $\mathcal{V} = \text{Null } A$

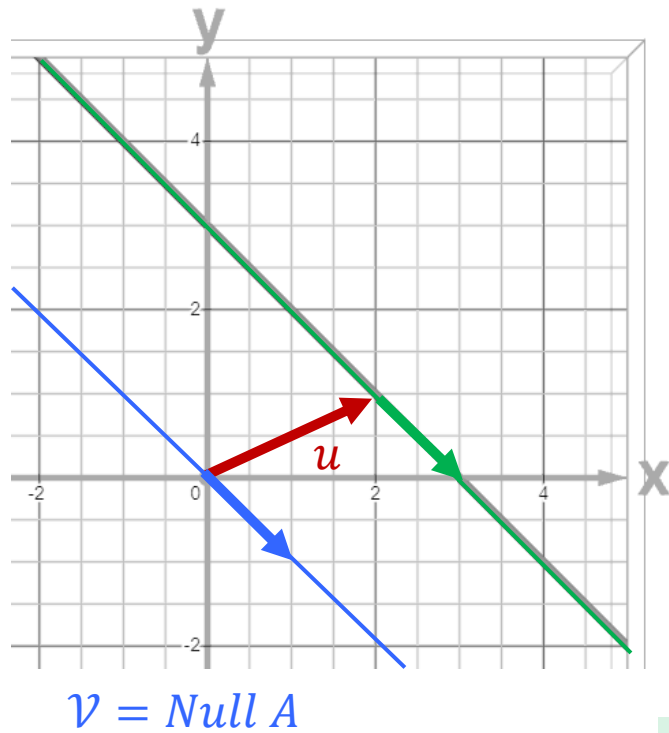
- ▶ If $\mathcal{V}$ is a trivial vector space then $\mathbf{u}_1$ is the only solution. (trivial $\rightarrow$ contains only the zero vector)
- ▶ If $\mathcal{V}$ is not trivial then $\mathbf{u}_1$ is *not* the only solution.

Corollary: $A * \mathbf{x} = \mathbf{b}$ has at most one solution iff $\text{Null } A$ is a trivial vector space.

We will later cover how we can determine whether a vector space is trivial



Null Space of a Matrix (VIII)



Linear Systems

$$x + y = 3$$



$$x + y = 0$$

Corresponding
homogeneous LE(s)



Solution set for homog. LS

$$\text{span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$



$$u + \mathcal{V}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \quad \text{Solution Set}$$

Matrix-Vector Equation

$$A \quad x \quad b \\ [1, 1] * [x, y]^T = [3]$$



$$[1, 1] * [x, y]^T = 0$$

Matrix-vector equation for
Null space of A



Nullspace of A

$$\text{Null } A = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$



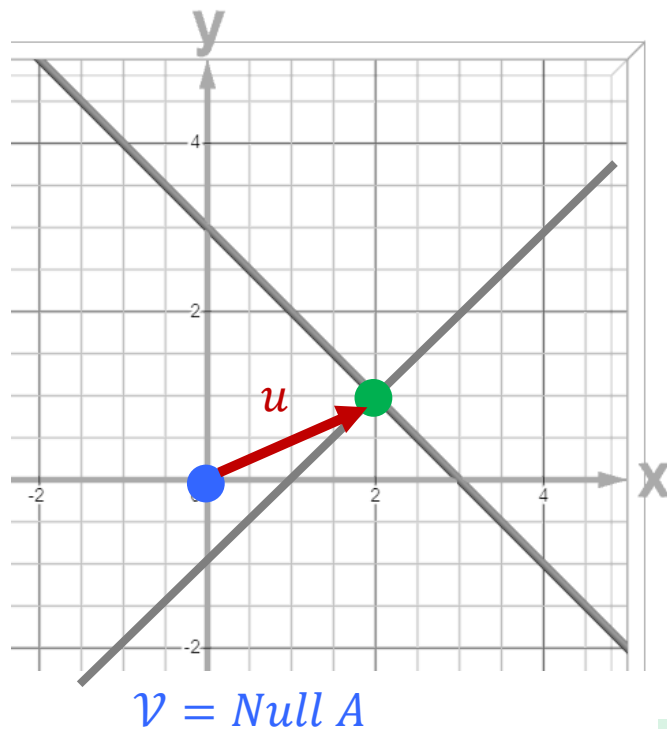
$$u + \text{Null } A$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \quad \text{Solution Set}$$



Null Space of a Matrix (X)

Let's add an equation!



Linear Systems

$$\begin{aligned} x + y &= 3 \\ x - y &= 1 \end{aligned}$$



$$\begin{aligned} x + y &= 0 \\ x - y &= 0 \end{aligned} \quad \begin{array}{l} \text{Corresponding} \\ \text{homogeneous LS} \end{array}$$



Solution set for homog. LS

$$\text{span} \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



$$u + \mathcal{V}$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{Solution Set}$$

Matrix-Vector Equation

$$A \quad x \quad b \\ \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} * [x, y]^T = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} * [x, y]^T = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} \text{Matrix-vector} \\ \text{equation for} \\ \text{Null space of } A \end{array}$$



Nullspace of A

$$\text{Null } A = \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{Trivial Null space}$$



$$u + \text{Null } A$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \text{Solution Set}$$



Null Space of a Matrix (XI)

Linear Systems

$$u + \mathcal{V}$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{Solution Set}$$

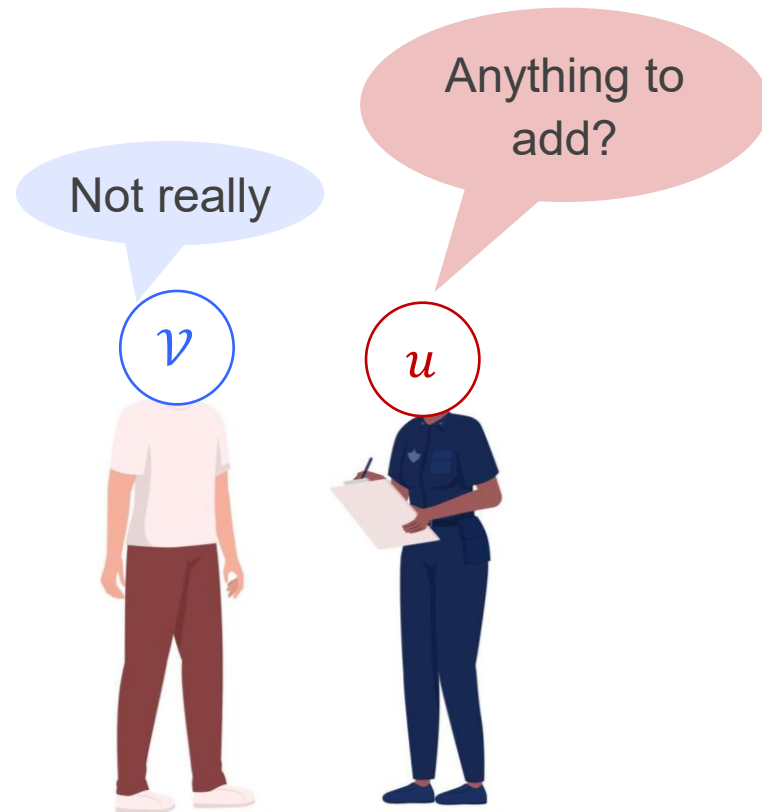
A system of linear equations has **at most one solution** iff the associated vector space $\mathcal{V}$ in the expression $u + \mathcal{V}$ contains **only the zero vector**.

Matrix-Vector Equation

$$u + \text{Null } A$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{Solution Set}$$

The matrix-vector equation $A * \mathbf{x} = \mathbf{b}$ has **at most one solution** iff $\text{Null } A$ is a **trivial vector space**

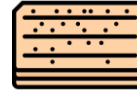




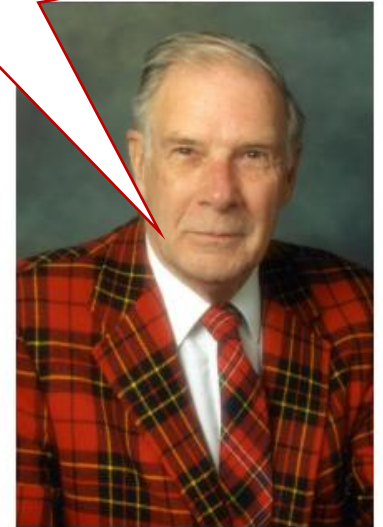
Error-Correcting Codes (I)

Hamming was tired of his computer restarting code when an error is present – but the computer provided no insight *where* the error was

- ▶ Originally inspired by errors in reading programs on punched cards
- ▶ Now used in WiFi, cell phones, communication with satellites and spacecraft, digital television, RAM, disk drives, flash memory, CDs, and DVDs



I'm getting too old for this *error*



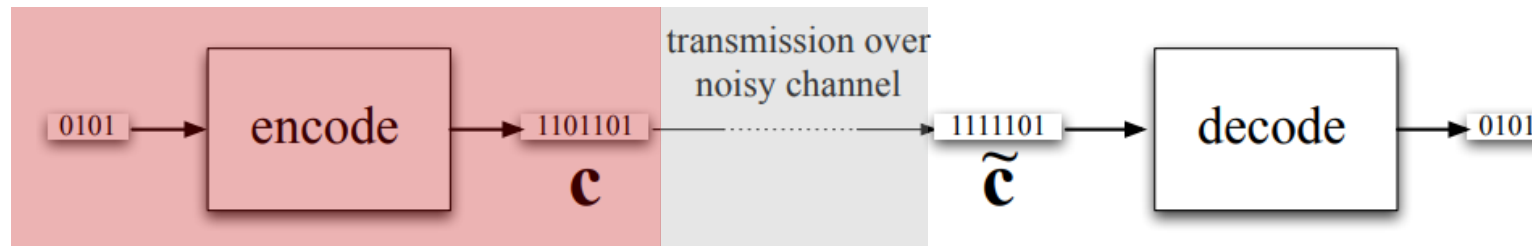
Richard Hamming

Hamming code is a linear binary block code:

- ▶ *linear* because it is based on linear algebra,
- ▶ *binary* because the input and output are assumed to be in binary, and
- ▶ *block* because the code involves a fixed-length sequence of bits.



Error-Correcting Codes (II)



To protect an 4-bit block:

- ▶ Sender *encodes* 4-bit block as a 7-bit block $\mathbf{c}$
- ▶ Sender transmits $\mathbf{c}$
- ▶ $\mathbf{c}$ passes through noisy channel—errors might be introduced.
- ▶ Receiver receives 7-bit block $\tilde{\mathbf{c}}$
- ▶ Receiver tries to figure out original 4-bit block

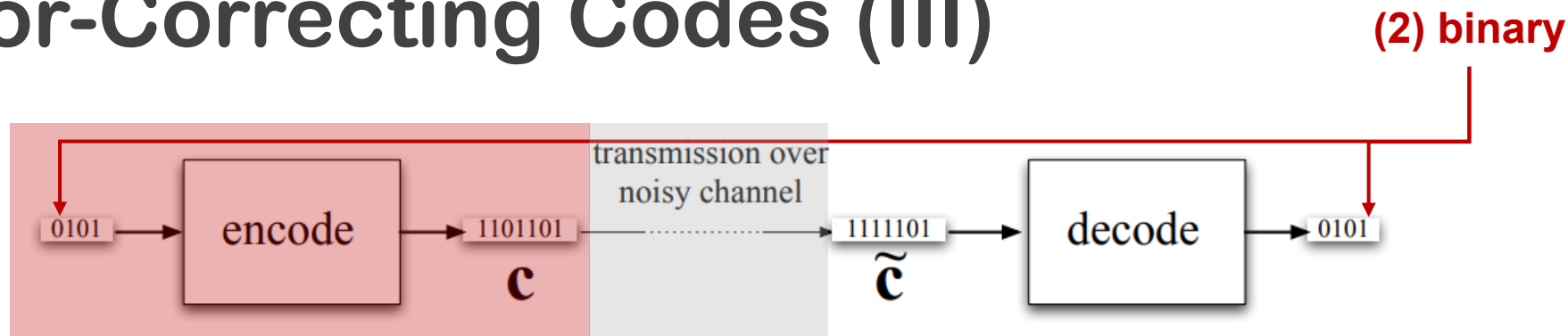
The 7-bit encodings are called *codewords*.

$\mathcal{C}$ = set of permitted codewords



[We focus on the LA aspect, not **how** to do the encoding and decoding]

Error-Correcting Codes (III)



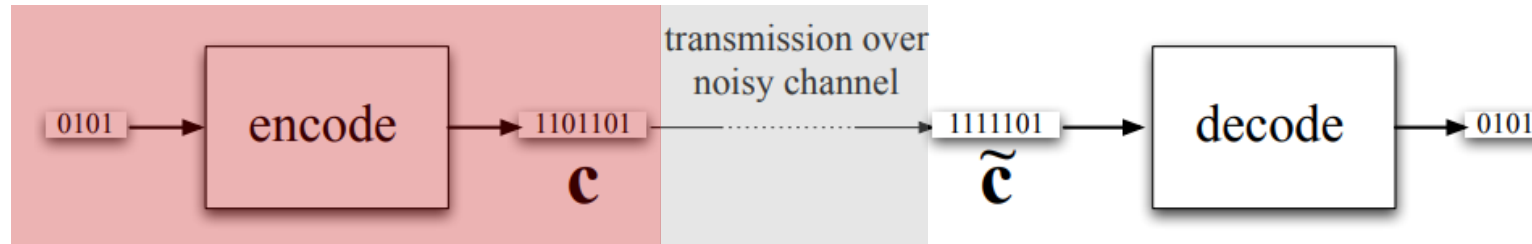
Hamming's first code is a *linear* code: **(1) linear** (any linear combination of codewords is also a codeword)

- ▶ Represent 4-bit and 7-bit blocks as 4-vectors and 7-vectors over $GF(2)$.
- ▶ 7-bit block received is $\tilde{\mathbf{c}} = \mathbf{c} + \mathbf{e}$ **(3) block**
- ▶ $\mathbf{e}$ has 1's in positions where noisy channel flipped a bit
($\mathbf{e}$ is the *error vector*)



[We focus on the LA aspect, not **how** to do the encoding and decoding]

Error-Correcting Codes (IV)



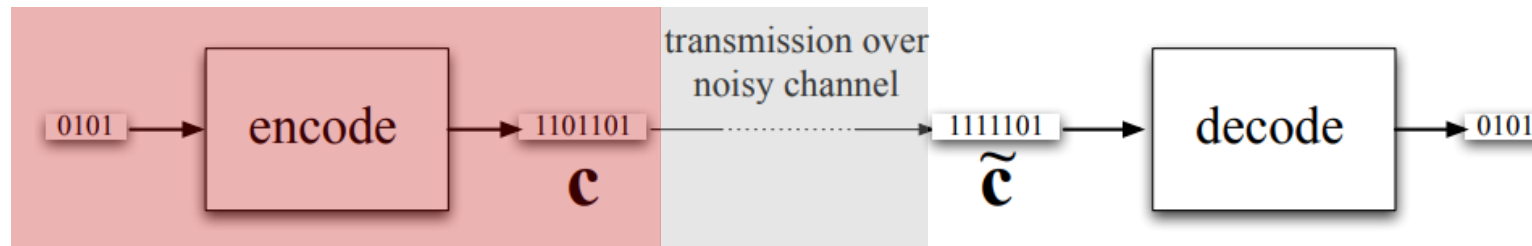
$$c = [1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 1]^T \quad \text{Codeword}$$

$$e = [0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0]^T \quad \text{Error (here: 1-bit error)}$$

$$\tilde{c} = c + e = [1 \ 1 \ 1 \ 1 \ 1 \ 0 \ 1]^T \quad \text{Received Codeword}$$



Error-Correcting Codes (V)



► **Key idea:** set $\mathcal{C}$ of codewords is the null space of a matrix H .

*Null Space of H is $\{\mathbf{c} : H * \mathbf{c} = \mathbf{0}\}$*

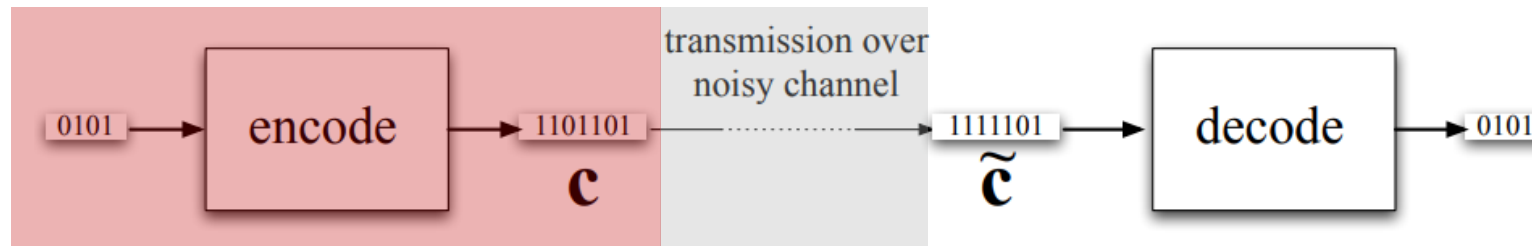
➡ $H * \mathbf{c} = \mathbf{0}$ Original message should lead to zero vector

Definition: Null space of a matrix A is $\{\mathbf{u} : \underline{A * \mathbf{u} = \mathbf{0}}\}$. Written $Null A$

Earlier in
this lecture



Error-Correcting Codes (VI)



Remember:
We want to know
(1) is there an error
and (2) where?

- ▶ **Key idea:** set $\mathcal{C}$ of codewords is the null space of a matrix H .

This makes Receiver's job easier:

- ▶ Receiver has $\tilde{\mathbf{c}}$, needs to figure out $\mathbf{e}$.
- ▶ Receiver multiplies $\tilde{\mathbf{c}}$ by H .

$\mathbf{c}$ in $\text{Null } H$

$$H * \tilde{\mathbf{c}} = H * (\mathbf{c} + \mathbf{e}) = H * \mathbf{c} + H * \mathbf{e} = \mathbf{0} + H * \mathbf{e} = H * \mathbf{e}$$

- ▶ Receiver must calculate $\mathbf{e}$ from the value of $H * \mathbf{e}$. *How?*



Error-Correcting Codes (VII)

In the Hamming code, the codewords are 7-vectors, and

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Notice anything special about the columns and their order?

- ▶ Suppose that the noisy channel introduces at most one bit error.
- ▶ Then $\mathbf{e}$ has only one 1.
- ▶ Can you determine the position of the bit error from the matrix-vector product $H * \mathbf{e}$?



Error-Correcting Codes (VIII)

(Remember: Matrix-vector Multiplication can be thought of as taking (a multiple) of the **columns** of the matrix

In the Hamming code, the codewords are 7-vectors, and

$$H \times e = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \times [0, 0, 1, 0, 0, 0, 0]^T = [0, 1, 1]$$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$

Example: Suppose $\mathbf{e}$ has a 1 in its third position, $\mathbf{e} = [0, 0, 1, 0, 0, 0, 0]^T$. **Third element has an error**

Then $H * \mathbf{e}$ is the third column of H , which is $[0, 1, 1]$.

As long as $\mathbf{e}$ has at most one bit error, the position of the bit can be determined from $H * \mathbf{e}$. This shows that the Hamming code allows the recipient to correct one-bit errors.



Error-Correcting Codes (IX)

In the Hamming code, the codewords are 7-vectors, and

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Quiz: Show that the Hamming code does not allow the recipient to correct two-bit errors: give two different error vectors, $\mathbf{e}_1$ and $\mathbf{e}_2$, each with at most two 1's, such that $H * \mathbf{e}_1 = H * \mathbf{e}_2$.



Error-Correcting Codes (X)

In the Hamming code, the codewords are 7-vectors, and

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Answer: There are many acceptable answers. For example, $\mathbf{e}_1 = [1, 1, 0, 0, 0, 0, 0]$ and $\mathbf{e}_2 = [0, 0, 1, 0, 0, 0, 0]$ or $\mathbf{e}_1 = [0, 0, 1, 0, 0, 1, 0]$ and $\mathbf{e}_2 = [0, 1, 0, 0, 0, 0, 1]$.

$$(1) H * [1, 1, 0, 0, 0, 0, 0]^T = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$(1) H * [0, 0, 1, 0, 0, 0, 0]^T = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Two different possible errors lead to the same outcome

$$(2) H * [0, 0, 1, 0, 0, 1, 0]^T = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$(2) H * [0, 1, 0, 0, 0, 0, 1]^T = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

... same here



We have M and use it to do $Mx = b$

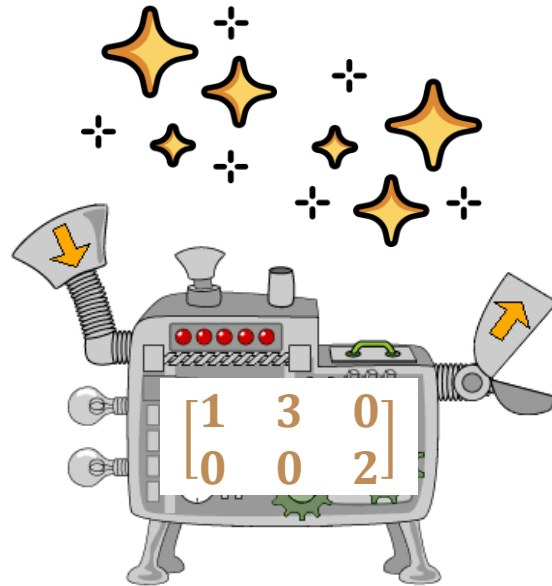
From Matrix to Function (I)

Now we study the relationship between a matrix M and the function $\mathbf{x} \mapsto M * \mathbf{x}$

What are the input (**domain**) and output (**co-domain**) sizes?

Domain-vector

$$x = [? \dots ?]^T$$



Codomain-vector

$$b = [? \dots ?]^T$$

Function $f(x) = Mx = b$

We know M ,
how large are
the **inputs** and
outputs?



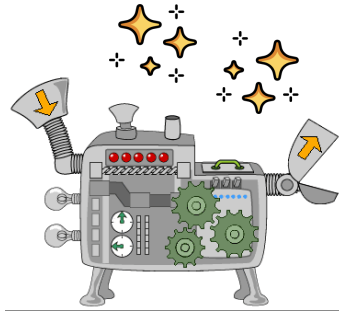


We have M and use it to do $Mx = b$

From Matrix to Function (II)

C-vector

$$x = [?, \dots, ?]^T$$



R-vector

$$b = [7, 6]^T$$

$$\text{Function } f(x) = Mx = b$$

Think of matrix-vector multiplication as a linear combination – when would this work?

$$\begin{matrix} M \\ \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{matrix} * [?, \dots, ?]^T \rightarrow \text{Number of components in inputs needs to match the number of columns in } M \rightarrow \begin{matrix} M \\ \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{matrix} * \begin{matrix} \text{Here:} \\ \text{3-vector} \\ [x_1, x_2, x_3]^T \end{matrix}$$

Now what is the **output size**?
Correspond to the number of rows

$$\begin{matrix} M \\ \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{matrix} * \begin{matrix} \text{3-vector} \\ [x_1, x_2, x_3]^T \end{matrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

2-by-3 M **2-vector**



EASY

We have M and use it to do $Mx = b$

From Matrix to Function (II)

$$\begin{matrix} M \\ \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{matrix} * [?, \dots, ?]^T \rightarrow \text{Number of components in inputs needs to match the number of columns in } M \rightarrow \begin{matrix} M \\ \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{matrix} * \begin{matrix} \text{Here:} \\ \text{3-vector} \\ [x_1, x_2, x_3]^T \end{matrix}$$

Now what is the **output size**?
Correspond to the number of rows

$$\begin{matrix} M \\ \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{matrix} * \begin{matrix} \text{3-vector} \\ [x_1, x_2, x_3]^T \end{matrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{matrix} \text{2-vector} \\ \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \end{matrix}$$

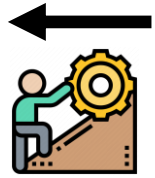
Domain and co-domain?

If M is an $R \times C$ matrix over $\mathbb{F}$ then

- ▶ domain of f is $\mathbb{F}^C$
- ▶ co-domain of f is $\mathbb{F}^R$

($\mathbb{F}$ to denote a field)

Inputs: C -vector (number of columns)
Outputs: R -vector (number of rows)



Something is happening, we know a (so far unknown) M is responsible

From Function to Matrix (I)

We know the inputs and the output of the function, but need to **find the matrix** that maps inputs to the outputs

Three possible problems:

b unknown
1) $Mx = b$ solve using matrix-vector multiplication to multiply Mx



You call this a problem?!

x unknown
2) $Mx = b$ solve e.g., using systems of linear equations

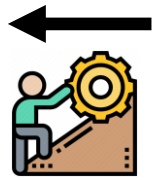


Not really a problem for me ...

M unknown
3) $Mx = b$ solve ... ok, no idea so far



Ok, that's a problem ...



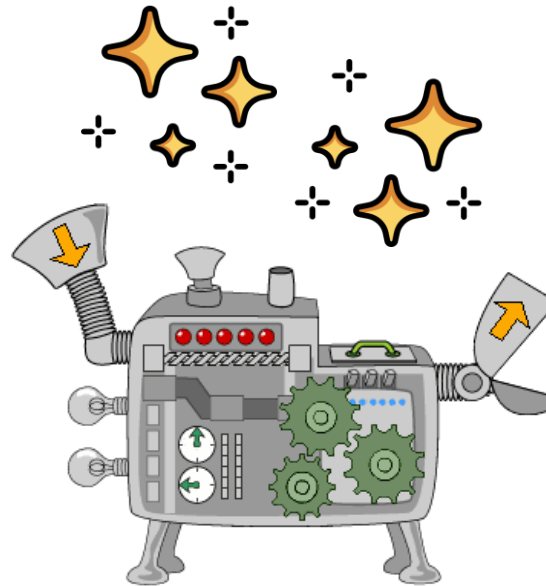
Something is happening, we know a (so far unknown) M is responsible

From Function to Matrix (II)

We use what we have learned before about the input and output sizes and how they match the matrix implementing the function

4-vector

$$x = [1, 2, 4, 2]^T$$



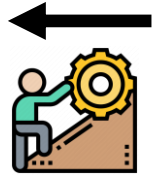
2-vector

$$b = [7, 6]^T$$

$$\text{Function } f(x) = Mx = b$$

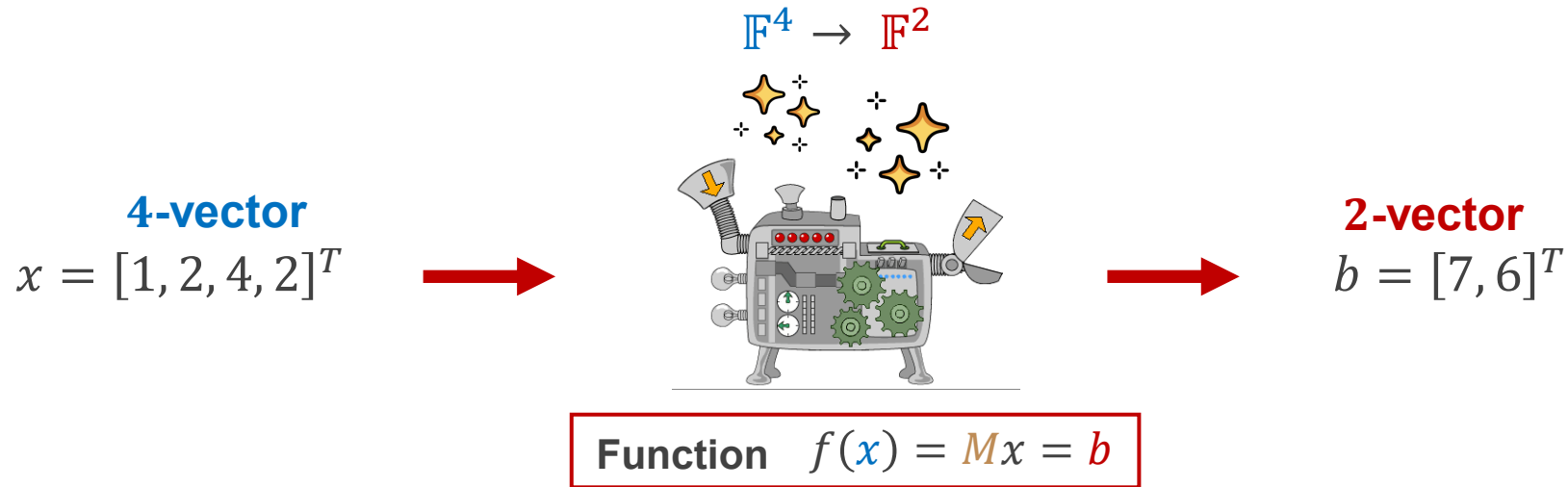
Before seeing M ,
how many rows
and columns does
it need to have?



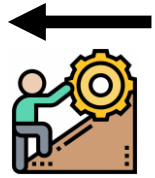


Something is happening, we know a (so far unknown) M is responsible

From Function to Matrix (III)

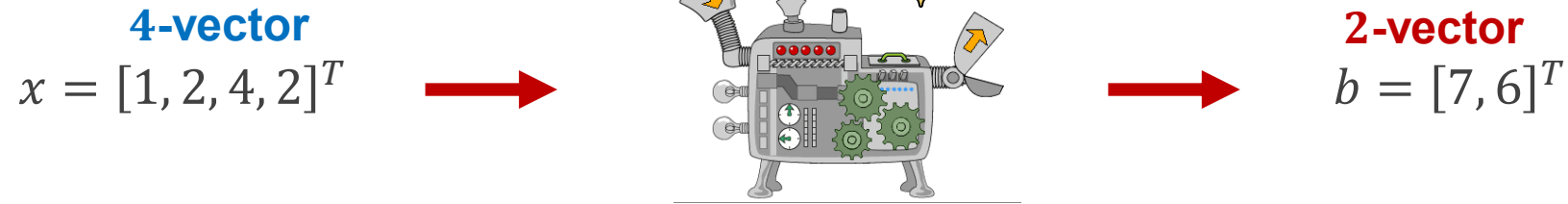


$$\begin{matrix} & M & & & \\ \left[\begin{array}{c|c|c} v_{1,1} & \dots & v_{1,?} \\ \dots & \dots & \dots \\ v_{?,1} & \dots & v_{?,?} \end{array} \right] & * & \begin{matrix} \text{4-vector} \\ [x_1, x_2, x_3, x_4]^T \end{matrix} & = & \begin{matrix} \text{2-vector} \\ \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \end{matrix} \end{matrix}$$



Something is happening, we know a (so far unknown) M is responsible

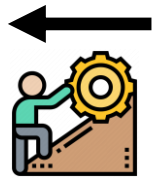
From Function to Matrix (III)



$$\begin{matrix} & M & & \text{4-vector} & \text{2-vector} \\ \left[\begin{array}{c|c|c|c} v_1 & \dots & v_? & \end{array} \right] & * & [x_1, x_2, x_3, x_4]^T & = & \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \end{matrix}$$

Four **columns** needed to match the inputs in $\mathbb{F}^4$

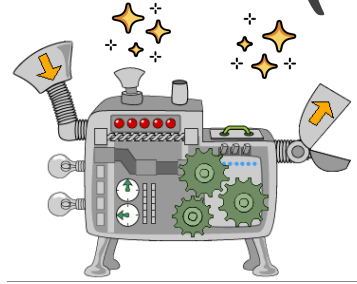
$$x_1 \begin{bmatrix} v_1 \end{bmatrix} + x_2 \begin{bmatrix} v_2 \end{bmatrix} + x_3 \begin{bmatrix} v_3 \end{bmatrix} + x_4 \begin{bmatrix} v_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \quad \text{2-vector}$$



Something is happening, we know a (so far unknown) M is responsible

From Function to Matrix (IV)

4-vector
 $x = [1, 2, 4, 2]^T$



2-vector
 $b = [7, 6]^T$

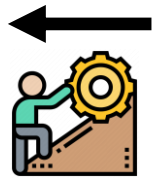
$$x_1 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + x_2 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + x_3 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + x_4 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Two rows needed to
match the outputs in $\mathbb{F}^2$

$$\begin{bmatrix} x_1 v_{1,1} \\ x_1 v_{2,1} \end{bmatrix} + \begin{bmatrix} x_2 v_{1,2} \\ x_2 v_{2,2} \end{bmatrix} + \begin{bmatrix} x_3 v_{1,3} \\ x_3 v_{2,3} \end{bmatrix} + \begin{bmatrix} x_4 v_{1,4} \\ x_4 v_{2,4} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$



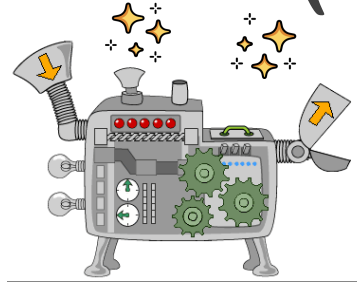
$$\begin{bmatrix} x_1 v_{1,1} + x_2 v_{1,2} + x_3 v_{1,3} + x_4 v_{1,4} \\ x_1 v_{2,1} + x_2 v_{2,2} + x_3 v_{2,3} + x_4 v_{2,4} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$



Something is happening, we know a (so far unknown) M is responsible

From Function to Matrix (V)

4-vector
 $x = [1, 2, 4, 2]^T$



2-vector
 $b = [7, 6]^T$

$$\begin{bmatrix} x_1 v_{1,1} + x_2 v_{1,2} + x_3 v_{1,3} + x_4 v_{1,4} \\ x_1 v_{2,1} + x_2 v_{2,2} + x_3 v_{2,3} + x_4 v_{2,4} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

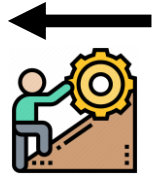
$$\begin{matrix} M(2\text{-by-}4) & \text{4-vector} & \text{2-vector} \\ \begin{bmatrix} v_{1,1} & \dots & v_{1,4} \\ \dots & \dots & \dots \\ v_{2,1} & \dots & v_{2,4} \end{bmatrix} & * [x_1, x_2, x_3, x_4]^T & = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \end{matrix}$$

In general: We have a function $f : \mathbb{F}^A \longrightarrow \mathbb{F}^B$

Now we know that M must be a $B \times A$ matrix....

We know the size M needs to be

... but what about its entries?



Something is happening, we know a (so far unknown) M is responsible

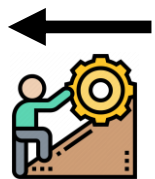
From Function to Matrix (VI)

► Write mystery matrix in terms of its columns: $M = \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix}$ $R \times C$ matrix

Linear Combination perspective of matrix-vector product

$$\begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix} \times [x_1, \dots, x_n]^T = x_1 \begin{bmatrix} v_1 \end{bmatrix} + \cdots + x_n \begin{bmatrix} v_n \end{bmatrix}$$

(The sum of the scalar multiples of the columns)



Something is happening, we know a (so far unknown) M is responsible

From Function to Matrix (IV)



Welcome back old friend,
standard generator



► Write mystery matrix in terms of its columns: $M = \left[\begin{array}{c|c|c} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{array} \right]$

► Use standard generators $\mathbf{e}_1 = [1, 0, \dots, 0, 0], \dots, \mathbf{e}_n = [0, \dots, 0, 1]$ with *linear-combinations* definition of matrix-vector multiplication:

$$f(\mathbf{e}_1) = \left[\begin{array}{c|c|c} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{array} \right] * [1, 0, \dots, 0, 0]^T = 1v_1 + 0v_2 \dots + 0v_n = v_1$$

Selects only **first** column

⋮

$$f(\mathbf{e}_n) = \left[\begin{array}{c|c|c} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{array} \right] * [0, 0, \dots, 0, 1]^T = 0v_1 + 0v_2 \dots + 1v_n = v_n$$

Selects only **n-th** column



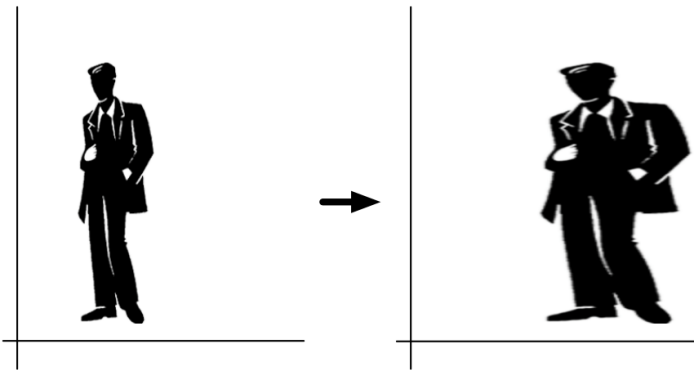
Matrix - Scaling (I)

Note: Not all axis are necessarily scaled the same way as with scalar multiplication

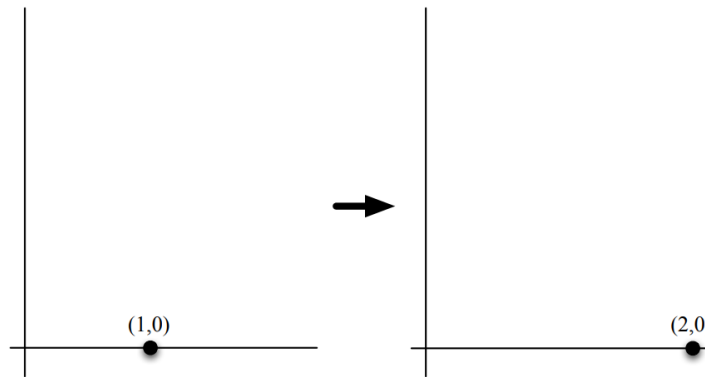
Define $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$ = stretching by two in horizontal direction

Assume $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = M * [x, y]^T$ for some matrix M .

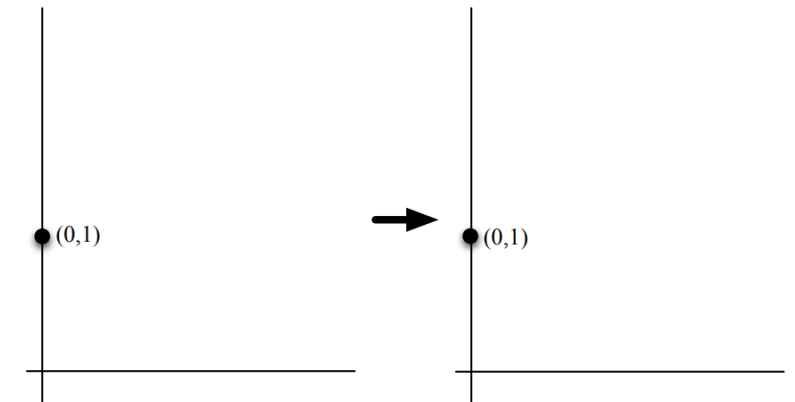
Therefore $M = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$



Stretched by 2 (for x)



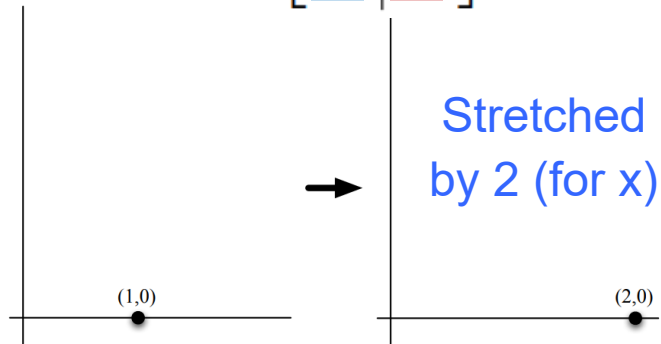
No change (for y)





Matrix - Scaling (II)

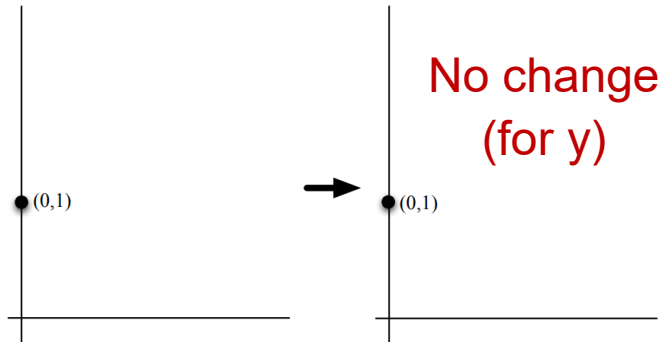
$$M = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$



Standard generator $e_1 = [1, 0]^T$ when multiplied with M only uses the **first column**, and, thus, takes on only M 's first column's values

$$M * e_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} * [1, 0]^T = 1 * \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 0 * \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

All other columns multiplied by zero → no impact



Same logic for $e_2 = [0, 1]^T$, which when multiplied with M only uses the **second column**, and, thus, takes on only M 's second column's values

$$M * e_2 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} * [0, 1]^T = 0 * \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 1 * \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

This thought ONLY works for the standard generators – but helps to understand what the matrix M does to them

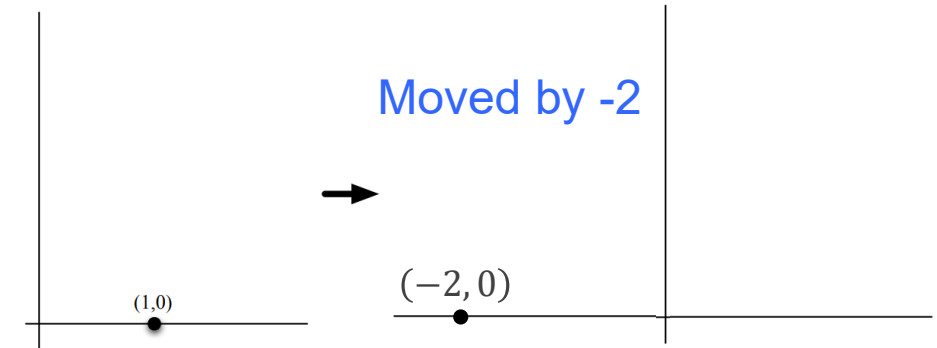


Matrix applied to Standard Generators (I)

To make sure we fully understand this, let's shortly do the reverse – let's think about what we want each standard generator to become, and create an M matrix accordingly

We want $e_1 = [1,0]^T$ to be $[-2,0]^T$ and $e_2 = [0,1]^T$ to be $[1,2]^T$

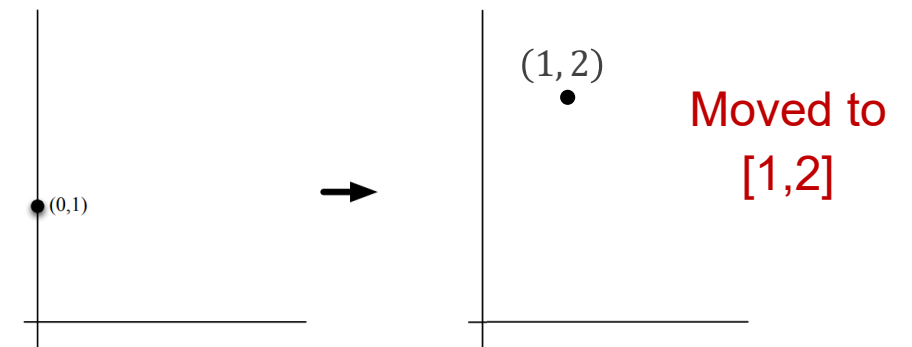
$$M = \begin{bmatrix} -2 & 1 \\ 0 & 2 \end{bmatrix}$$



Short double-check

$$M * e_1 = \begin{bmatrix} -2 & 1 \\ 0 & 2 \end{bmatrix} * [1,0]^T = 1 * \begin{bmatrix} -2 \\ 0 \end{bmatrix} + 0 * \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

$$M * e_2 = \begin{bmatrix} -2 & 1 \\ 0 & 2 \end{bmatrix} * [0,1]^T = 0 * \begin{bmatrix} -2 \\ 0 \end{bmatrix} + 1 * \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$





Matrix applied to Standard Generators (II)

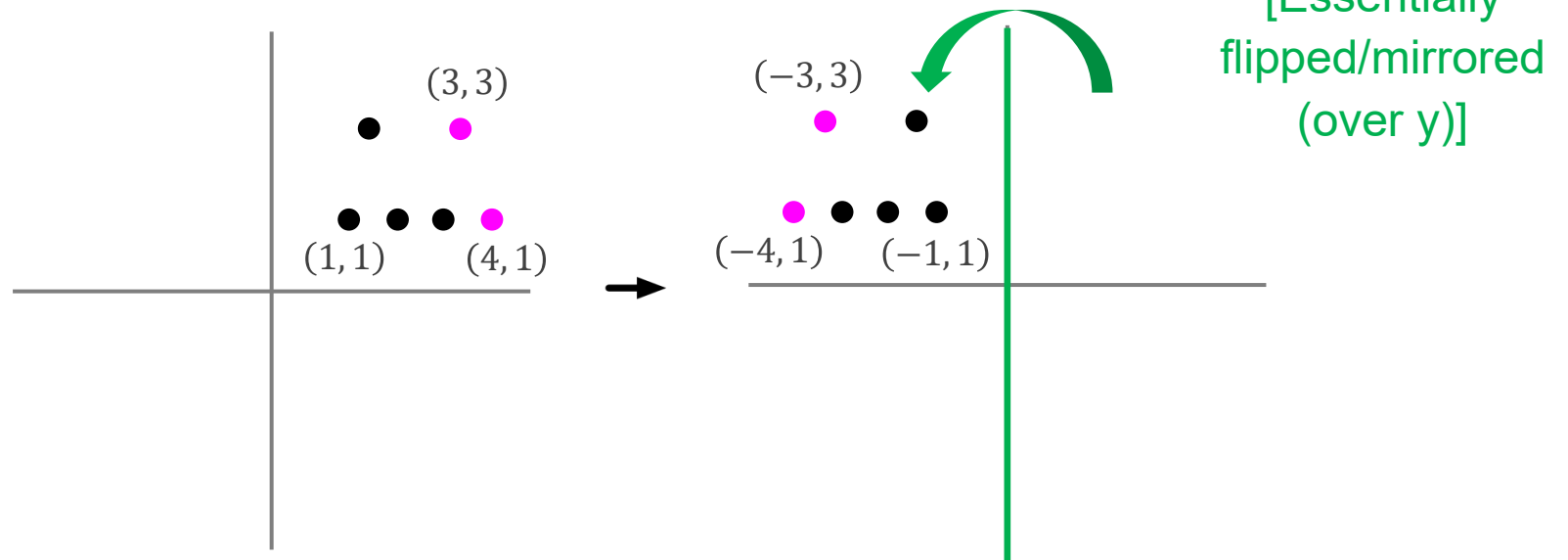
Define $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$ = First dimension/axis changes its sign

Assume $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = M * [x, y]^T$ for some matrix M .

- ▶ We know $f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ because the sign is reversed
- ▶ We know $f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ because no change in vertical direction.

Therefore $M = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

Sign for x reversed (only for x)





Matrix applied to Standard Generators (III)

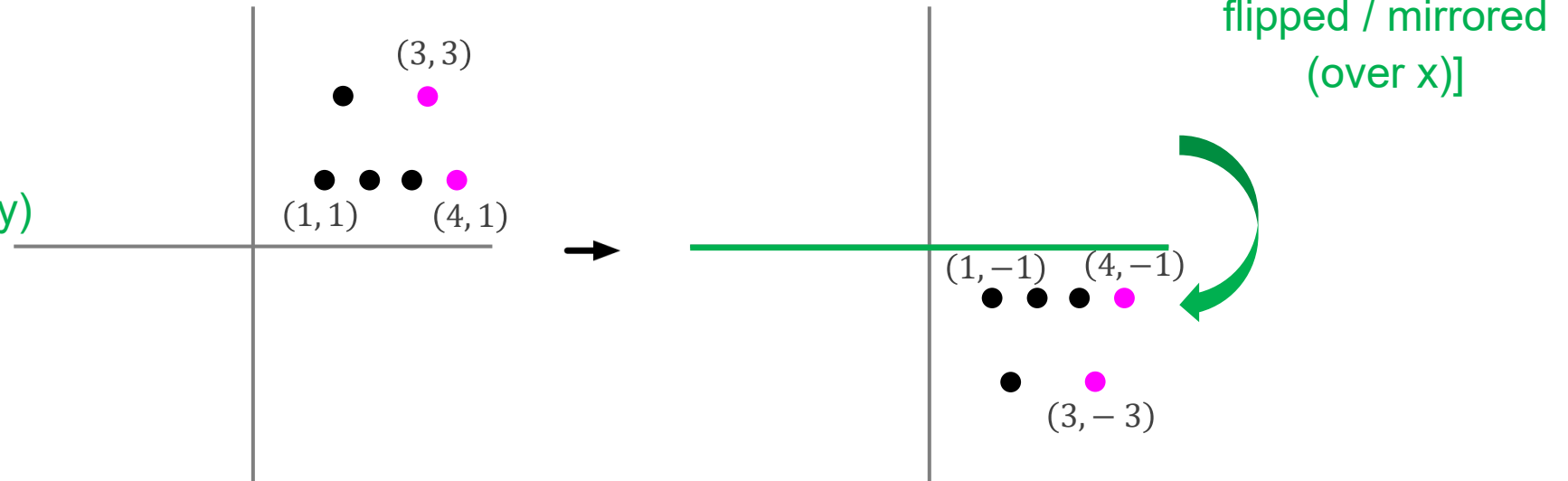
Define $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$ = Second dimension/axis changes its sign

Assume $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = M * [x, y]^T$ for some matrix M .

- ▶ We know $f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
- ▶ We know $f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

Therefore $M = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Sign for y reversed (only for y)





Matrix applied to Standard Generators (IV)

Define $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$ = Both dimensions change their sign

Assume $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = M * [x, y]^T$ for some matrix M .

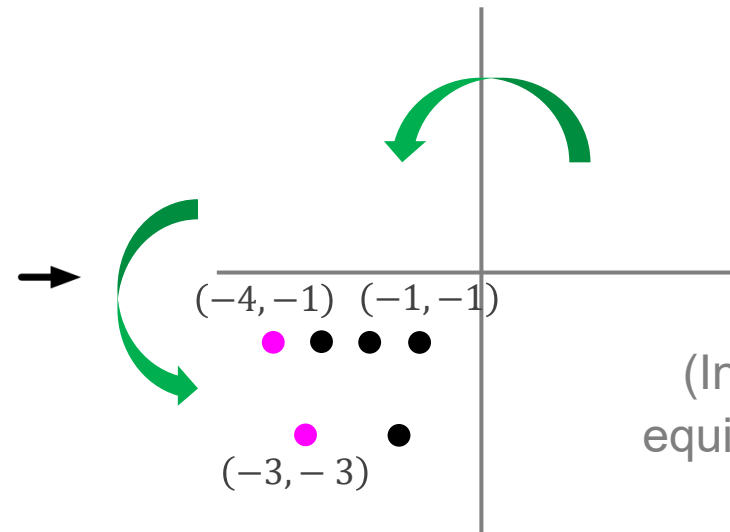
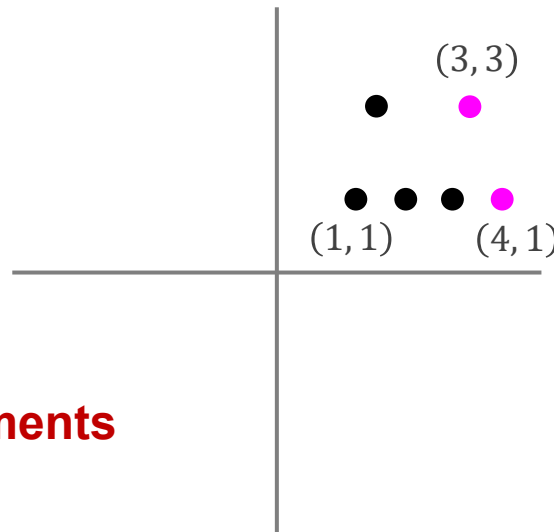
► We know $f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

► We know $f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

Therefore $M = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

Signs reversed (for x and y)

**So far, we only adjusted elements
on the diagonal**



[Essentially
flipped /
mirrored twice
(over x and y)]

(Interestingly, is also
equivalent to performing
a 180° rotation)



Matrix applied to Standard Generators (V)

Define $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$ = Exchange the x and y location (NOT rotation – see highlighted pink points)

Assume $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = M * [x, y]^T$ for some matrix M .

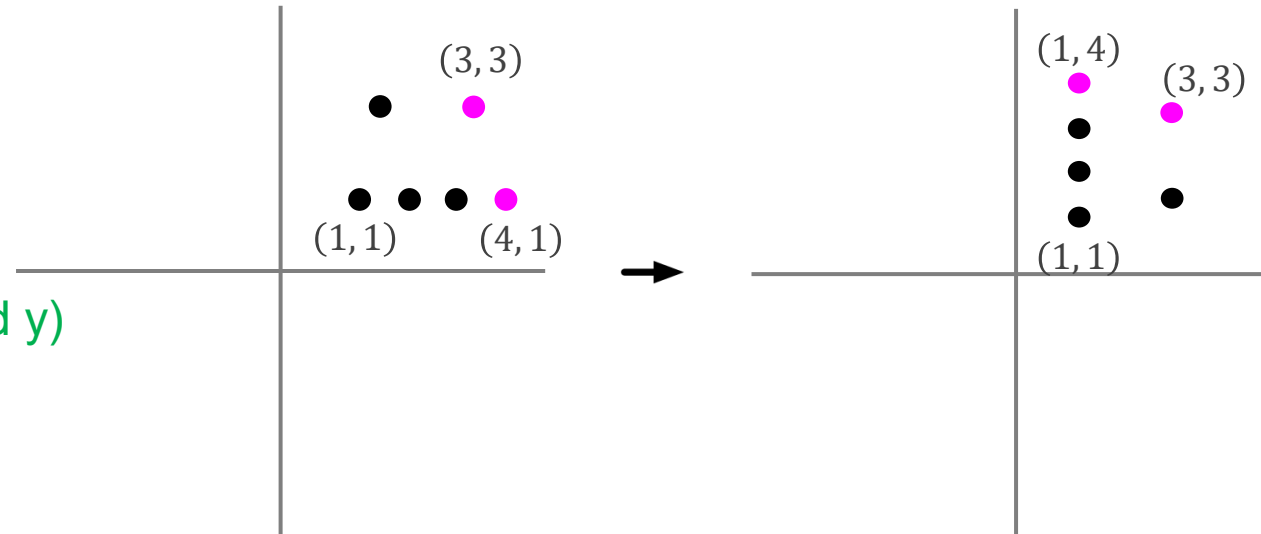
► We know $f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

► We know $f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Changes on the off-diagonal

Therefore $M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Exchanged axis location (for x and y)





Remember: **Key idea:** set $\mathcal{C}$ of codewords is the null space of a matrix H .

Exercise: Error Correcting Codes

You are given the following H matrix, encoded messages c_1 and c_2 , and received messages $\tilde{c}_3$

- 1) Check if the encoded codewords c_1 and c_2 are actually in $\text{Null } H$ (the Null Space of matrix H)

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

- 2) How would the error e vector look like if there was an error in the 3rd, 5th and 6th bit / entry?
- 3) Add the error from the previous task to c_1 to create the received message $\tilde{c}_1$. Check if the error would be detected. Explain why / why not?
- 4) Given the received message $\tilde{c}_3$, determine if there is an error and if yes, where. Show work.

$$c_1 = [1, 0, 0, 1, 1, 0, 0] \quad \tilde{c}_3 = [1, 1, 1, 1, 0, 0, 0]$$

$$c_2 = [0, 1, 0, 0, 1, 1, 0]$$

Solution

Exercise: E.C. Codes

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \begin{array}{l} c_1 = [1, 0, 0, 1, 1, 0, 0] \\ c_2 = [0, 1, 0, 0, 1, 1, 0] \\ \tilde{c}_3 = [1, 1, 1, 1, 0, 0, 0] \end{array}$$

a) If a vector x is in $\text{Null } A$, then $Ax = 0$

$$H \times c_1 = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \times [1, 0, 0, 1, 1, 0, 0] = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$H \times c_2 = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \times [0, 1, 0, 0, 1, 1, 0] = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Result interpretation / answer: Thus, c_1 is in $\text{Null } A$, but c_2 is not (so it is not a valid codeword)

b) $[0, 0, 1, 0, 1, 1, 0]$ (error in the 3rd, 5th and 6th bit / entry)

Solution

Exercise: E.C. Codes

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \begin{array}{l} \mathbf{c}_1 = [1, 0, 0, 1, 1, 0, 0] \\ \mathbf{c}_2 = [0, 1, 0, 0, 1, 1, 0] \\ \tilde{\mathbf{c}}_3 = [1, 1, 1, 1, 0, 0, 0] \end{array}$$

$$\text{c) } \tilde{\mathbf{c}}_1 = \mathbf{c}_1 + \mathbf{e}_1 = [1, 0, 0, 1, 1, 0, 0] + [0, 0, 1, 0, 1, 1, 0] = [1, 0, 1, 1, 0, 1, 0]$$

$$H \times \tilde{\mathbf{c}}_1 = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \times [1, 0, 1, 1, 0, 1, 0] = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The corrupted message is also in $\text{Null } H$ so we would incorrectly assume there is no error (and not detect any error). This is not surprising as only single-bit errors can be detected reliably

$$\text{d) } H \times \tilde{\mathbf{c}}_3 = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \times [1, 1, 1, 1, 0, 0, 0] = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

The received message $\tilde{\mathbf{c}}_3$ is NOT in $\text{Null } H$ so we conclude that there is an error. The only single-bit error that could be responsible is $[1, 0, 0]$, which is equivalent to the fourth column in H .



Some Key Take-aways (Lecture 8)

The **null space for a matrix** A ($Null A$) is the vector that multiplied with the matrix gives the zero vector and is a **vector space** ($V1, V2, V3$)

If a vector u is a solution to a linear system (expressed via matrix-vector multiplication: $Ax = b$) then the solution set is $u + Null A$

} Link to homogeneous linear systems

Error correcting codes introduced by Hamming matrix may help to determine errors and their position (in sequences of bits)

} Application using the Null Space

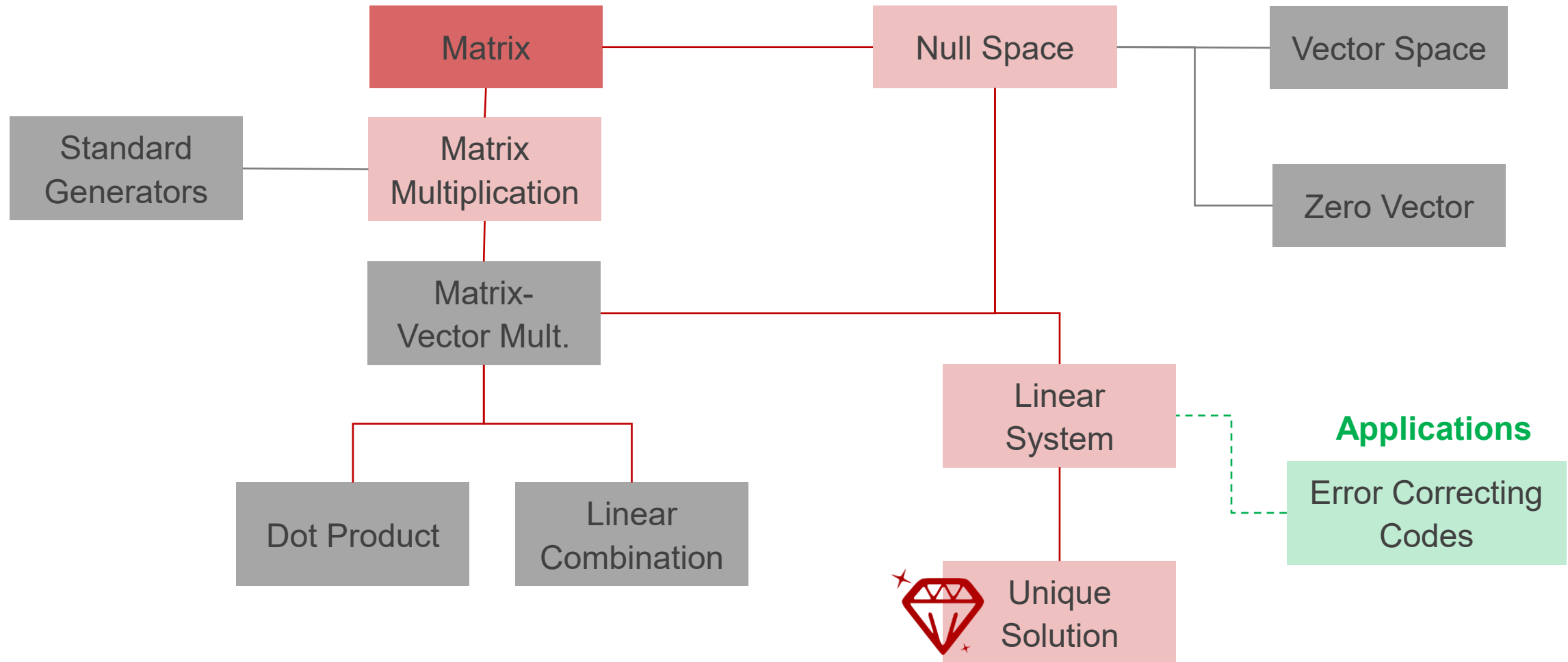
It is challenging from a **(linear) function** to understand what is the matrix M that is multiplied with the known input x to obtain the known output b .

The **standard generators** may assist us in understanding what the individual columns of the (unknown) matrix M does to them



Lecture 8 - Summary

Concepts



Previous

Applications



Thank You For Your Attention!



**Accomplishment unlocked:
Lecture 8 Completed**



APPENDIX



Matrices and their Functions

Now we study the relationship between a matrix M and the function $\mathbf{x} \mapsto M * \mathbf{x}$

- ▶ *Easy:* Going from a matrix M to the function $\mathbf{x} \mapsto M * \mathbf{x}$
- ▶ *A little harder:* Going from the function $\mathbf{x} \mapsto M * \mathbf{x}$ to the matrix M .

In studying this relationship, we come up with the fundamental notion of a linear function.

So, having matrix M to create an output $M * x$ is easy, but having an output $M * x$ we know was created using some matrix M and **inferring the matrix M that generated it**, is hard





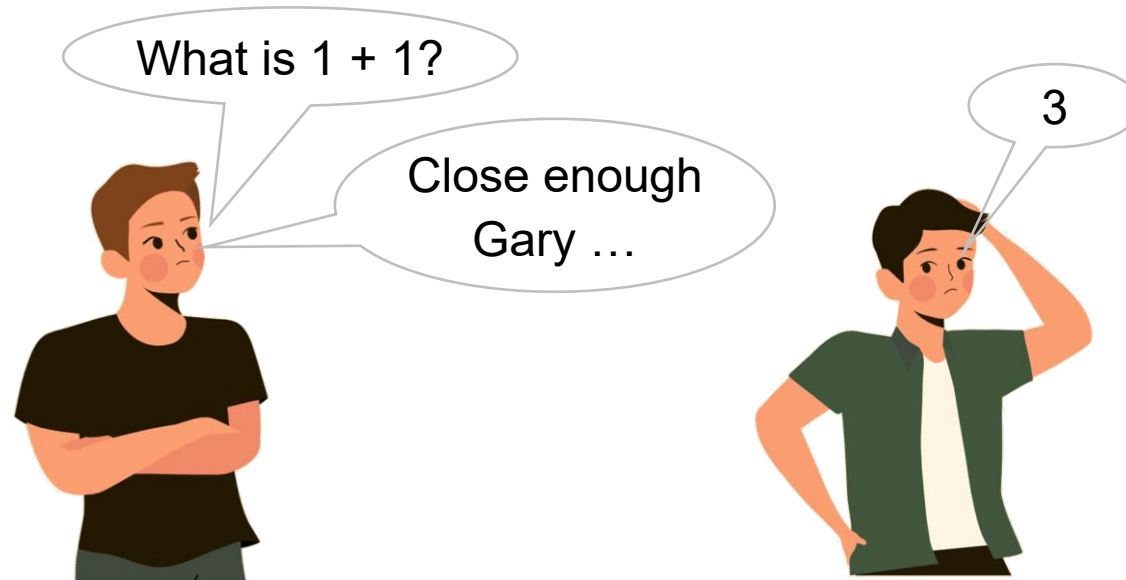
Solver Module (I)

For arithmetic over $\mathbb{R}$, Python uses floats, so round-off errors occur:

```
>>> 10.0**16 + 1 == 10.0**16
```

! True

Consequently algorithms such as that used in `solve(A, b)` do not find exactly correct solutions.





Solver Module (II)

To see if solution $\mathbf{u}$ obtained is a reasonable solution to $A * \mathbf{x} = \mathbf{b}$, see if the vector $\mathbf{b} - A * \mathbf{u}$ has entries that are close to zero:

```
>>> A = listlist2mat([[1,3],[5,7]])
>>> u = solve(A, b)
>>> b - A*u
Vec({0, 1},{0: -4.440892098500626e-16, 1: -8.881784197001252e-16})
```

The vector $\mathbf{b} - A * \mathbf{u}$ is called the *residual*. Easy way to test if entries of the residual are close to zero: compute the dot-product of the residual with itself:

```
>>> res = b - A*u
>>> res * res
9.860761315262648e-31
```

That is pretty small ($\rightarrow$ very close to zero)



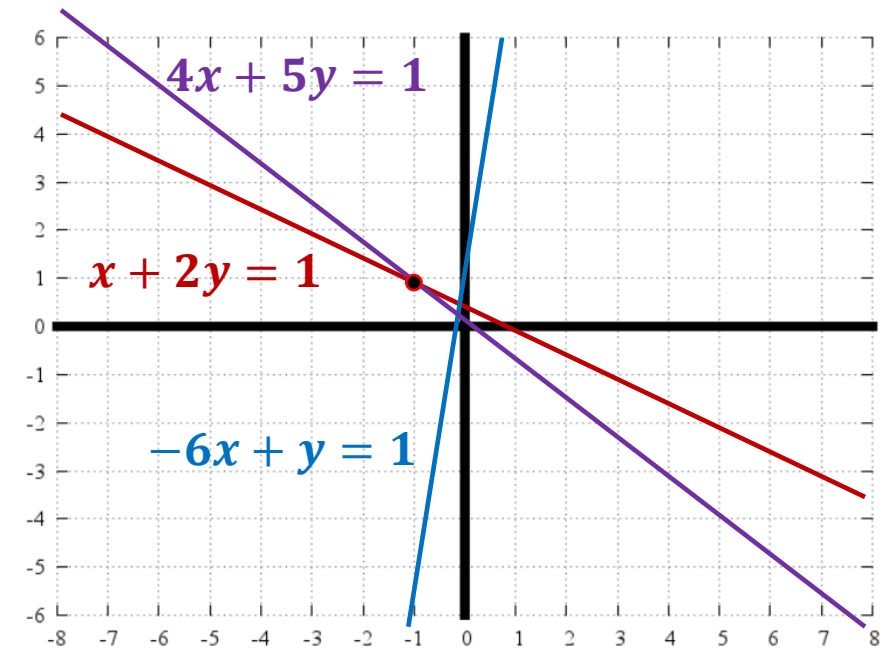
Solver Module (III)

For some matrix-vector equations $A * \mathbf{x} = \mathbf{b}$, there is **no solution**.

In this case, the vector returned by `solve(A, b)` gives rise to a **largeish residual**:

```
>>> A = listlist2mat([[1,2],[4,5],[-6,1]])
>>> b = list2vec([1,1,1])
>>> u = solve(A, b)
>>> res = b - A*u
>>> res * res
0.24287856071964012
```

That is not that small!





Solver Module (IV)

Some matrix-vector equations are *ill-conditioned*, which can prevent an algorithm using floats from getting even approximate solutions, even when solutions exists:

```
>>> A = listlist2mat([[1e20,1],[1,0]])  
>>> b = list2vec([1,1])  
>>> u = solve(A, b)  
>>> b - A*u  
Vec({0, 1},{0: 0.0, 1: 1.0}) That is not small at all!
```

We will not study conditioning in this course.